

Project Concept Note: HireSense AI

Vibe Coding Masterclass Series — Capstone Project Submission
Application Name: HireSense AI — Intelligent Career Readiness & Interview Co-Pilot
Developer: Student Submission
Live Application URL: https://hiresense-ai.vercel.app
Backend API URL: https://hiresense-ai-backend-ndq3.onrender.com/api/v1

1. Executive Summary & Problem Statement

1.1 The Challenge

Preparing for technical and HR job interviews is often stressful, fragmented, and lacks real-time, objective feedback. Job candidates consistently encounter three major barriers:

- Unrealistic Practice:** Static interview prep lists do not simulate dynamic, multi-turn technical conversations.
- ATS Resume Rejection:** Resumes are frequently filtered out by Automated Tracking Systems (ATS) due to formatting mismatches or missing role-specific keywords.
- Unstructured Learning Paths:** Candidates lack personalized, week-by-week actionable roadmaps tailored to their specific skill gaps and targeted roles.

1.2 The Solution

HireSense AI is an end-to-end, next-generation career readiness co-pilot powered by **Google Gemini AI** (`gemini-1.5-flash`). It provides: 1. **Real-time AI Mock Interviews:** Multi-turn practice with progressive token streaming. 2. **ATS Resume Calibration:** Section-by-section auditing and keyword matching. 3. **Holistic Performance Dashboards:** Multi-category scoring across Technical Depth, Communication, and Problem Solving. 4. **Custom Weekly Roadmaps:** Actionable skill-gap learning pathways tailored to candidate timelines.

2. Target Audience & Primary Use Cases

Target Audience	Primary Use Case	Key Benefit Received
Students & Fresh Graduates	Preparing for entry-level tech & HR interviews	Instant mock practice without needing human interviewers
Experienced Engineers	Transitioning into Senior/Lead/Specialized roles	Deep technical metric evaluation and targeted trade-off analysis
Career Switchers	Up-skilling for software engineering / data roles	Customized week-by-week learning pathway with project milestones

3. LLM Model & API Architecture

- **AI SDK:** Official `google-genai` Python SDK (`v2.12.1`)
- **Primary LLM Model:** Google Gemini `gemini-1.5-flash`
- **Streaming Transport:** Server-Sent Events (SSE) streaming via FastAPI `StreamingResponse`
- **Security & Privacy Guarantee:** Zero client-side API key exposure; all credentials stored in server environment variables; zero persistent storage of candidate resumes.

4. Key Application Features

Feature 1: Interactive AI Mock Interview Suite

- Dynamic role, experience level, and interview type selection (Technical, Behavioral, HR).
- Multi-turn conversation tracking with real-time SSE token streaming for progressive typing animations.

Feature 2: Live Answer Evaluation & Grading

- Detailed feedback per question: Score (0.0–10.0), key strengths, identified gaps, and rewritten "Gold Standard" answers.

Feature 3: Holistic Performance Dashboard

- Session-wide analytics across Technical Depth, Communication Clarity, and Problem Solving.
- Overall readiness grade (e.g. "Industry Ready", "Needs Practice") with actionable next steps.

Feature 4: Resume ATS Scanner & Calibrator

- Instant plain-text resume auditing against targeted job titles.
- Section scores, ATS readability metrics, weakness identification, and tailoring recommendations.

Feature 5: Week-by-Week Career Roadmap Generator

- Custom learning path creation matching current candidate skills against target role requirements.
- Hands-on weekly tasks, practical projects, and verification milestones.

5. Expected Outcomes & User Experience

Candidates experience a responsive web application featuring dark-mode aesthetic styling, micro-animations, and real-time streaming feedback.

- **5x Faster Interview Preparation:** Instant, repeatable practice sessions.
- **Measurable ATS Resume Score Improvements:** Clear formatting and keyword guidance.
- **Actionable Learning Pathways:** Structured weekly tasks eliminate career pivot ambiguity.